CS1 Week 11 Annotated

Frequency-domain Specifications, Loop shaping

Recap Last Week

Quiz 1

What does a single point in the polar plot of a system $G(j\omega)$ represent?

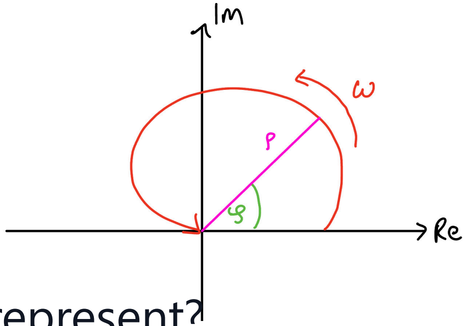
A. The open-loop gain $\boldsymbol{k}$ at frequency $\boldsymbol{\omega}$

B. The complex value $\boldsymbol{G(j\omega)}$, consisting of magnitude and phase

C. Only the phase of $\boldsymbol{G(j\omega)}$, plotted as an angle on the imaginary axis

D. Only the magnitude of $\boldsymbol{G(j\omega)}$, plotted as a radius

Quiz 1



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C. Only the phase of $G(j\omega)$, plotted as an angle on the imaginary axis

D. Only the magnitude of $G(j\omega)$, plotted as a radius

Quiz 2

Which of the following statements are correct?

A. With Nyquist we can determine closed-loop stability only by looking at the open-loop

B. $N = Z - P$, with $N =$ encirclements about $-1/k$; $Z =$ unstable open-loop zeros; $P =$ unstable open-loop poles

C. $N = Z - P$, with $N =$ encirclements about $-1/k$; $Z =$ unstable closed-loop poles; $P =$ unstable open-loop poles

D. Nyquist is a more robust way to determine stability of a closed-loop system

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D. Nyquist is a more robust way to determine stability of a closed-loop system

Quiz 3

Which of the following statements about Nyquist plots are correct?

A. They are symmetric w.r.t. the real axis

B. The frequency is implicit in the Nyquist plot

C. We can add Nyquist Plots of different TFs on top of each other

D. They start at $\omega = \infty$ and end at $\omega = 0$

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Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	Polar plots, The Nyquist condition	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

1. Gain & Phase Margin
2. Frequecy-Domain Specifications
3. Loop Shaping

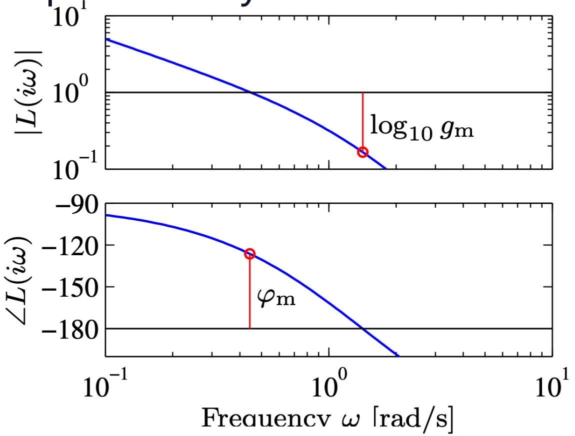
1. Gain & Phase Margin

Gain & Phase Margin

$$N = Z - P$$

- These margins are only valid for open-loop stable systems! ($P = 0$) $\longrightarrow$ for stability we need $N = 0$
- Phase & Gain Margin define "how close" we are to closed loop instability

(i.e. encircling the -1 point)

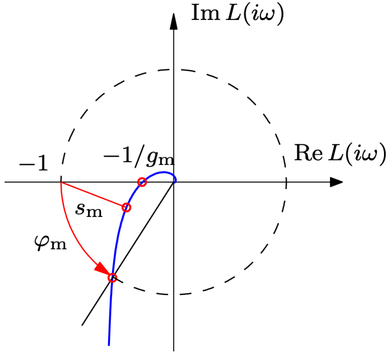


1. Gain Margin: $g_m = \frac{1}{|L(j\omega_{pc})|}$

How much can we change the gain at $-180°$ until we hit -1?

2. Phase Margin: $\varphi_m = 180° + \angle L(j\omega_c)$

How much can we change the phase at $0dB$ until we reach -1?

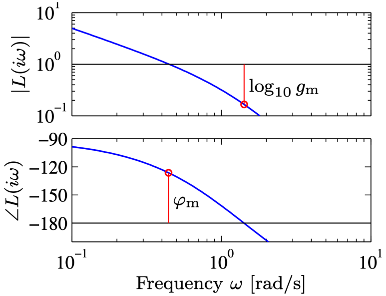


3. Bandwith:

Frequency where magnitude equals $\frac{1}{\sqrt{2}} = -3dB \approx \omega_{gc}$

measures how fast CL-System can react

Exam Problem



1. Gain Margin:
How much can we change the gain at $-180°$ until we hit -1?

2. Phase Margin:
How much can we change the phase at 0dB until we reach -1?

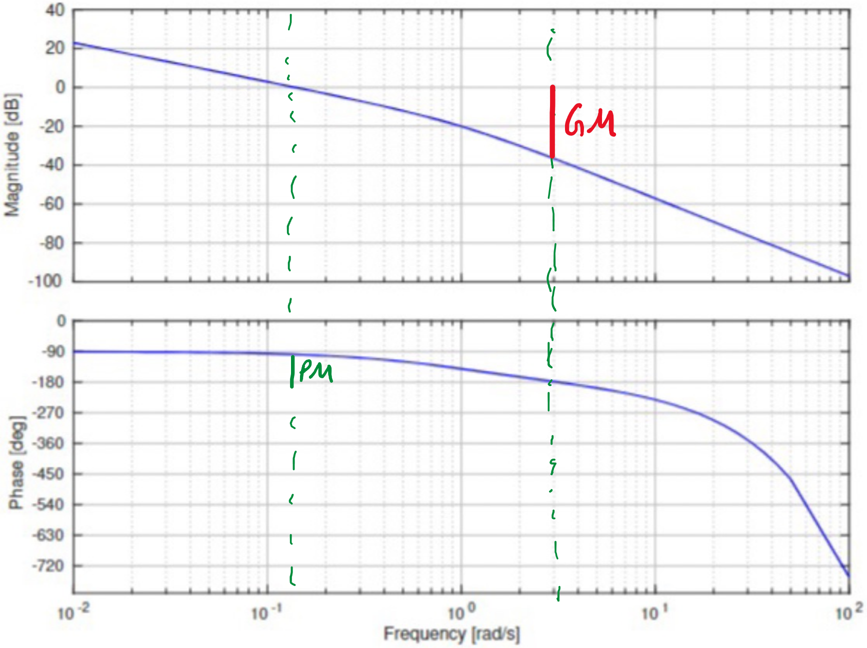


Figure 14: Bode plot of L.

Q29 **(0.5 Points)** Mark the correct answer.
The phase margin PM for closed-loop stability is best described by,

- A $PM \approx -83°$
- B $PM \approx 80°$
- C $PM \approx 47°$
- D $PM \approx -35°$

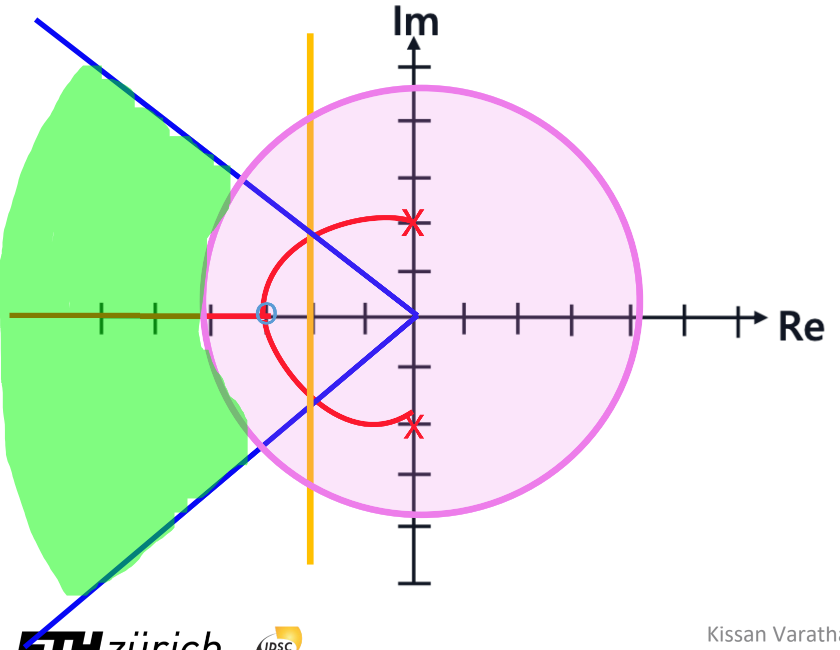
Q30 **(0.5 Points)** Mark the correct answer.
The gain margin GM for closed-loop stability is best described by,

- A $GM \approx -37$ dB
- B $GM \approx 82$ dB
- C $GM \approx 37$ dB
- D $GM \approx -21$ dB

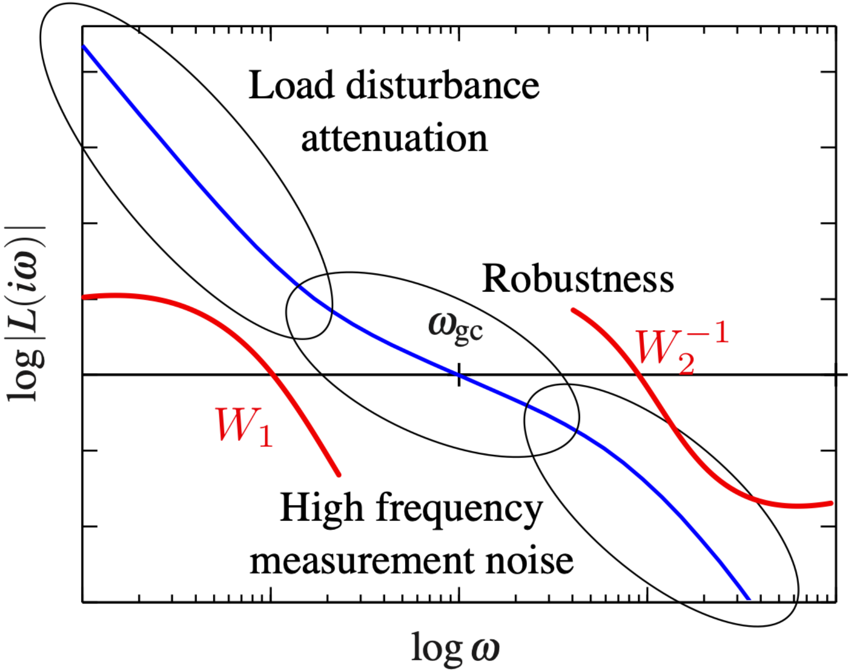
2. Frequency-domain Specifications

Overview

Three weeks ago, we looked at the **time-domain specifications** which looked like this:

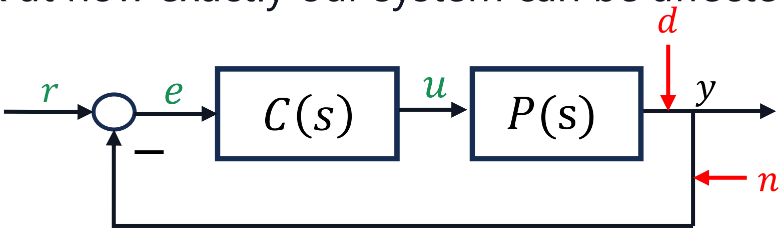


Today, we're going to look at the **frequency-domain specifications** which look like this:



Noise & Disturbance

Let's take a closer look at how exactly our system can be affected:



The effect of d and n on the system can be seen from the corresponding transfer functions:

- $d \longrightarrow e$: $\quad S(s) = \frac{1}{1+C(s)P(s)}$
- $n \longrightarrow e$: $\quad T(s) = \frac{C(s)P(s)}{1+C(s)P(s)}$

- The effect of both n and d on e should be as small as possible
- That means, both $S(s)$ and $T(s)$ should be as small as possible

There's one small issue…

- $d \longrightarrow e$: $S(s) = \frac{1}{1+C(s)P(s)}$
- $n \longrightarrow e$: $T(s) = \frac{C(s)P(s)}{1+C(s)P(s)}$

(Complementary) Sensitivity Functions

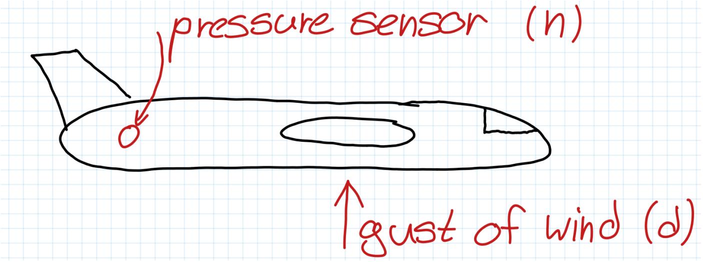
$$S(s) + T(s) = \frac{1}{1 + C(s)P(s)} + \frac{C(s)P(s)}{1 + C(s)P(s)} = 1$$

$\longrightarrow$ They always add up to one!

- This means that we can't cancel both noise and disturbances…

Thankfully, they have a property we can make use of:

- Noise usually occur at high frequencies
- Disturbances usually occur at low frequencies



Frequency-domain Specifications

So the workaround is to require:

1) $|S(s)| \ll 1$ for small ω
$\longrightarrow$ disturbance rejection

2) $|T(s)| \ll 1$ for large ω
$\longrightarrow$ noise rejection

How can we guarantee these constraints?

1) $S(s) = \frac{1}{1+L(s)} \quad \Rightarrow \quad S(s)$ small for small ω

We enforce the inequality using a constraint function $W_1(s)$: $|S(s)||W_1(s)| < 1$

for some $W_1(s)$ that is large at low ω

The equation can be rewritten as:

$$\left|\frac{1}{1+L(s)}\right| |W_1(s)| < 1 \Rightarrow |1+L(s)| > |W_1(s)|$$

$$\approx |L(s)| > |W_1(s)|$$

2) $T(s) = \frac{L(s)}{1+L(s)} \quad \Rightarrow \quad T(s)$ small at high ω

Again using a constraint function $W_2(s)$:

$$|T(s)||W_2(s)| < 1$$

for some $W_2(s)$ that is large at high ω

The equation can be rewritten as:

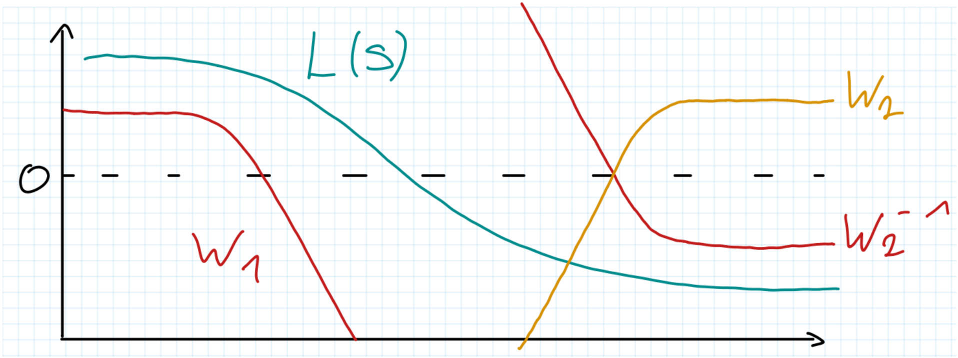
$$\left|\frac{L(s)}{1+L(s)}\right| |W_2(s)| < 1 \Rightarrow |L(s)|\,|W_2(s)| < 1$$

$$\approx |L(s)| < |W_2(s)|^{-1}$$

Visualization

$$|L(s)| > |W_1(s)|$$
$$|L(s)| < |W_2(s)|^{-1}$$

What exactly are these $W(s)$? :

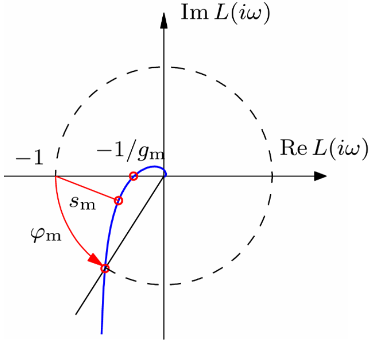


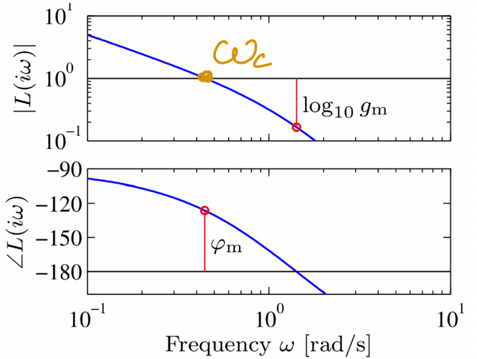
- The W's are like obstacles for the open-loop TF
- There's *one more important condition* for the closed-loop: $T(s) = \frac{L(s)}{1+L(s)}$

⟶ system becomes unstable for $L(s) = -1$

⟶ We could also see this in the Bode-plot!

Bode Plot

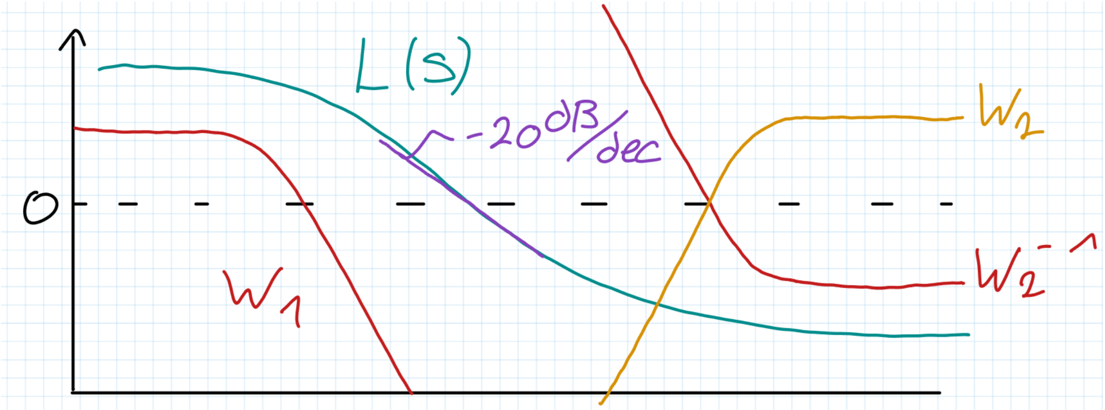




Another condition:

The slope of the magnitude plot is related to the phase!

- To ensure enough phase margin, the slope of the magnitude plot at the crossover frequency ω_c should be not steeper than $-20dB/dec$



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Dynamic Compensator

- All of these conditions for stability and performance make sense
- But our plant $P(s)$ just behaves however it behaves, we cannot change the physics

How do we ensure that the open loop $L(s)$ looks how we want?

$\Rightarrow$ Choose a controller $C(s)$ s.t. $L(s)$ does what we want!

Example: Given $P(s) = 0.01\,\frac{(s+10)^2}{s(s^2+2s+2)}$

Find a $C(s)$ such that the following conditions are respected:

- $W_1 = 100\text{dB} \;\; for\ \omega < 0.1\text{rad/s}$
- $W_2^{-1} = 0.05\text{dB}\ for\ \omega > 100\text{rad/s}$
- $\omega_c = 10\text{rad/s}$
- 45° phase margin at ω_c

Dynamic Compensator

$$P(s) = 0.01\frac{(s+10)^2}{s(s^2+2s+2)}$$

- $W_1 = 100\text{dB}\ for\ \omega < 0.1\text{rad/s}$
- $W_2^{-1} = 0.05\text{dB}\ for\ \omega > 100\text{rad/s}$
- $\omega_c = 10\text{rad/s}$
- 45° phase margin at ω_c

- A good open-loop suggestion would be:

$$L_{\text{des}} = 10 \cdot \frac{1}{s\left(\frac{s}{50}+1\right)}$$

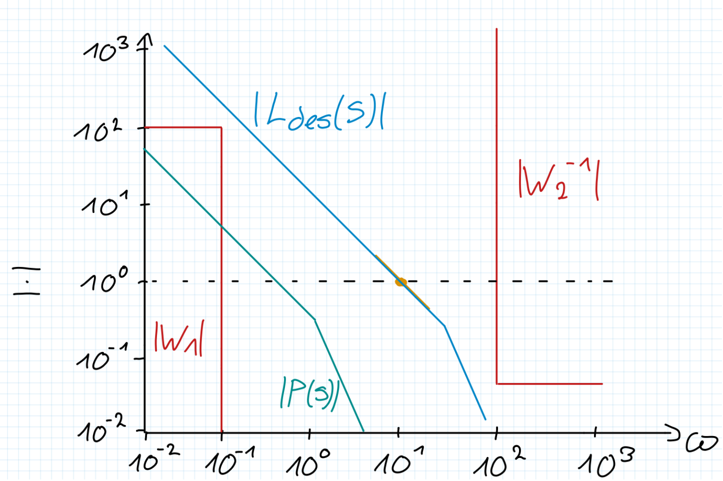
- Now we reverse engineer:

$$C(s) = \frac{L_{des}(s)}{P(s)} = 10 \cdot \frac{s^2+2s+2}{(s/50+1)(s/10+1)^2}$$

What we just did was a "brute force" approach…

- $C(s)$ may be unnecessarily complicated
- **Only works for open-loop stable, minimum-phase systems!**

otherwise, we would cancel unstable poles



A more elegant approach is called Loop Shaping…

Exam Question

dB	decimal
20	10
40	100
60	1 000
80	10 000
100	100 000

$$|L(s)| > |W_1(s)|$$
$$|L(s)| < |W_2(s)|^{-1}$$

Box 3: Questions 31, 32

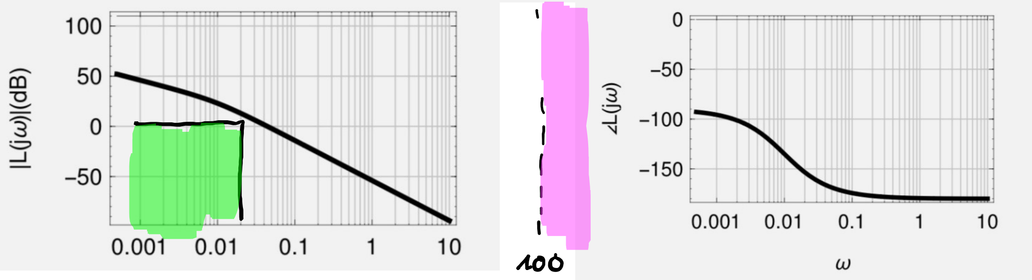
You are faced with the task of designing a controller $C(s)$ for the plant:

$$P(s) = \frac{1}{500\left(s^2 + \frac{s}{100}\right)}$$

and decide to use loop-shaping to solve the problem. You face the following constraints:

1. • Disturbances at frequencies $\omega \leq 0.01$ should be reduced by at least $60dB$.
2. • Noise which occurs at frequencies $\omega \geq 100$ should be attenuated by at least $100dB$.

The Bode plot of $P(s)$ is displayed below:



Question 31 *Choose the correct answer. (1 Point)*
How can the disturbance rejection and noise reduction specifications be represented in the Bode plot?

- [A] $|L(j\omega)| \approx 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- [B] $|L(j\omega)| > 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- [C] $|L(j\omega)| > 60dB$ for $\omega < 0.01$ and $|L(j\omega)| < -100dB$ for $\omega > 100$
- [D] $|L(j\omega)| < 60dB$ for $\omega < 0.01$ and $|L(j\omega)| > -100dB$ for $\omega > 100$
- [E] $|L(j\omega)| < 60dB$ for $\omega < 0.01$ and $|L(j\omega)| < -100dB$ for $\omega > 100$

2. $|T(s)| < -100dB$

$\left|\frac{L(s)}{\underbrace{1+L(s)}_{\approx 1}}\right| < -100dB \rightarrow |L(s)| < -100dB$

1. $|S(s)| < -60dB \rightarrow \left|\frac{1}{\underbrace{1+L(s)}_{\approx L(s)}}\right| < -60dB \rightarrow \frac{1}{|L(s)|} < -60dB$

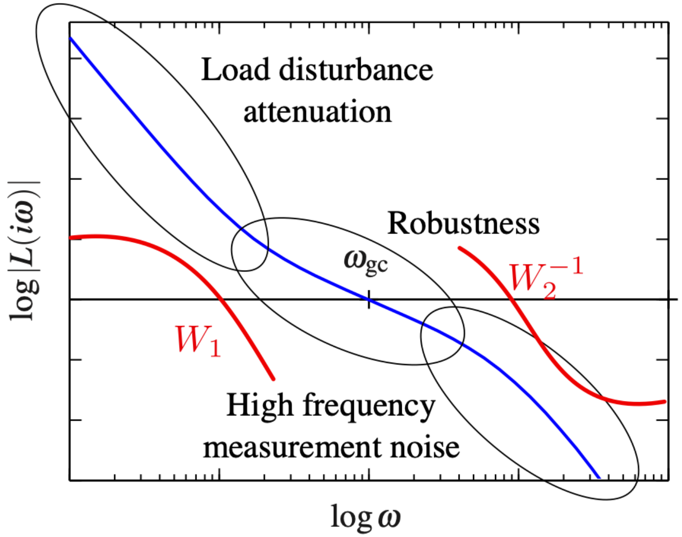
$\hookrightarrow |L(s)| > 60dB$

3. Loop Shaping

Loop Shaping

A better way to design our Compensator $C(s)$ is to use the following elements, s.t. $L(s)$ passes the obstacle course:

- Gain: k
- Integrator: $\frac{1}{s}$
- Lead: $\frac{s/a+1}{s/b+1}, 0 < a < b$
- Lag: $\frac{s/a+1}{s/b+1}, 0 < b < a$



⇒ We can then add each element in series how we want, i.e. $C(s) = C_1(s) \cdot C_2(s) \cdots C_n(s)$

⇒ But what is the effect of Lead/Lag-elements?

Lead-Element

- In essence, Lead-Lag elements are a combination of a stable pole & mnm. phase zero
- We use them to manipulate the frequency response

$$C_{lead} = \frac{s/a + 1}{s/b + 1}, 0 < a < b$$

Effects:

- The magnitude at high frequencies is **increased** by b/a. Low frequencies are not affected.
- The slope of the plot is increased by $+20dB/dec$ between a and b.

Magnitude response

- The phase is **increased** around $\sqrt{ab}$ by up to 90°. The maximum phase increase is:

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Phase response

Bode Plot of a Lead Element

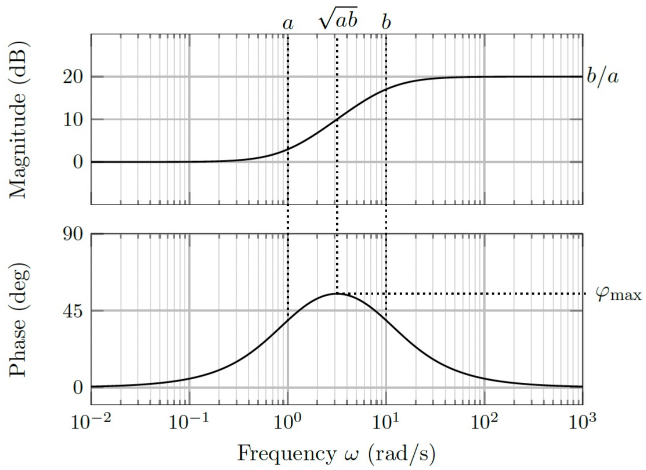


Figure 7.14: Bode Plot of a lead element with $a = 1$ and $b = 10$

Lag-Element

- A lag compensator is in many ways an inverse of the lead compensator

$$C_{lag} = \frac{s/a+1}{s/b+1}, 0 < b < a$$

Effects:

- The magnitude at high frequencies is **decreased** by b/a. Low frequencies are not affected.
- The slope of the plot is decreased by $-20\ dB/dec$ between b and a.

Magnitude response

- The phase is **decreased** around $\sqrt{ab}$ by up to 90°. The maximum phase decrease is:

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Phase response

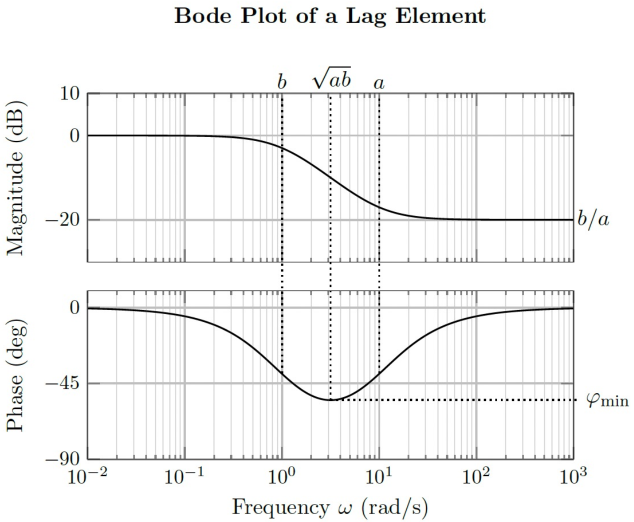


Figure 7.15: Bode Plot of a lag element with $b = 1$ and $a = 10$

Example

Hints 1 & 2:

- nmn-phase

higher ω_{gc} → faster rise time

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5\frac{(1-\frac{s}{200})}{(1+\frac{s}{20})(1+\frac{s}{4})}$$

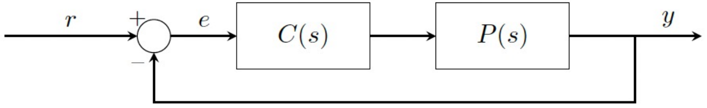


Figure 3: Standard feedback architecture.

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45°.

a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

$z_1 = 200 \Rightarrow \omega_c = 100$ //

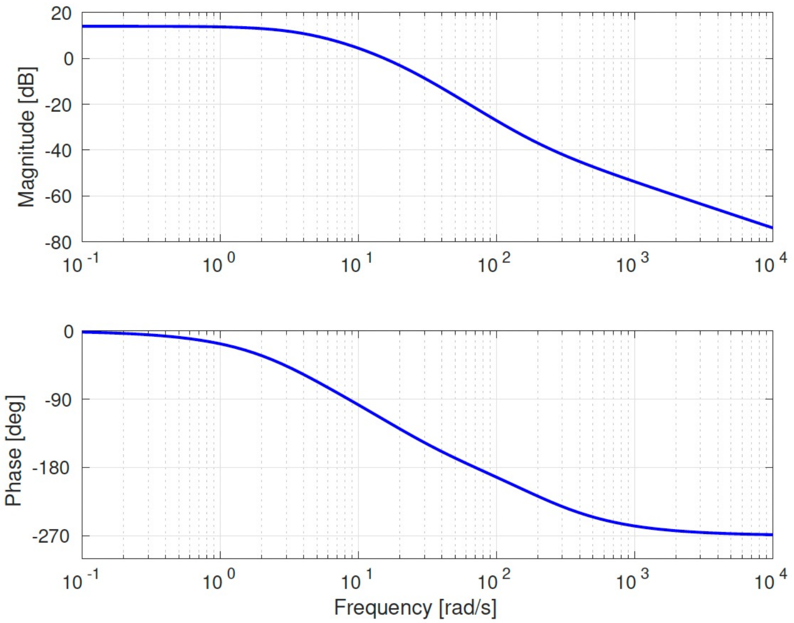


Figure 5: Bode plot for P(s).

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

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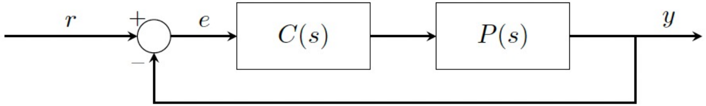

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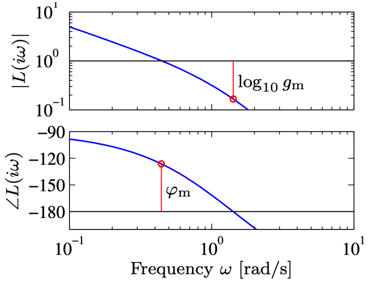

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3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45°.

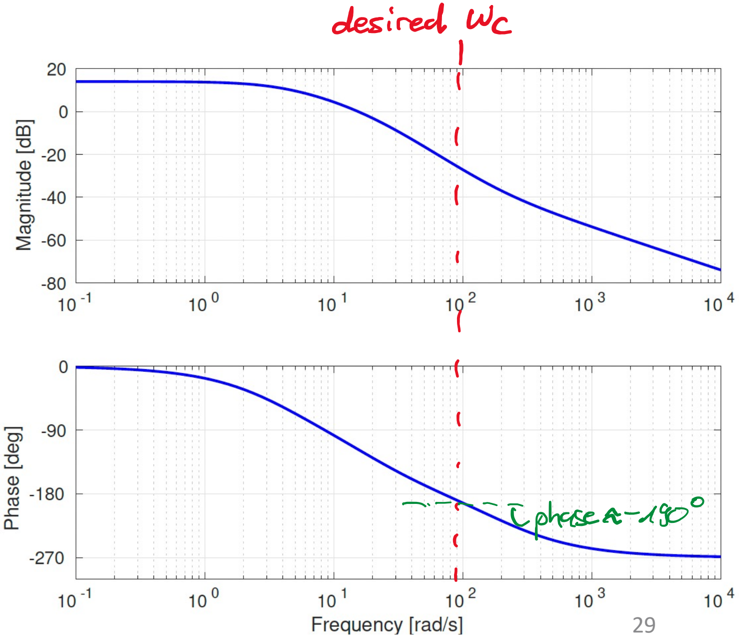
$\omega_c = 10^2$

a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

b) Determine the form of compensator required to achieve the desired crossover frequency and phase margin.

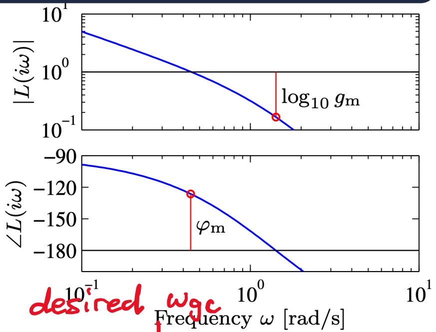
1) current ω_c smaller than desired ω_c
↳ need increase in magnitude

2) we need to increase phase to fulfill PM requirements



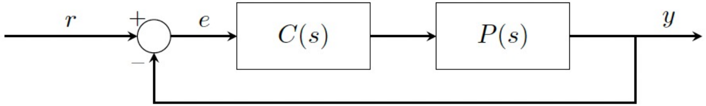
Example

Hints: Use your result from (a) to determine the phase of $P(j\omega)$ at $\omega = \omega_c$ and hence, the desired phase shift.



Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5\frac{(1 - \frac{s}{200})}{(1 + \frac{s}{20})(1 + \frac{s}{4})}$$



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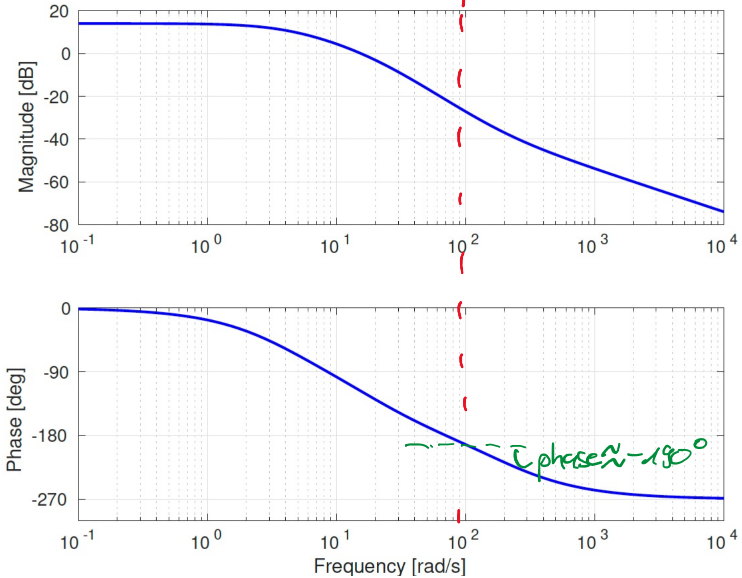
$\omega_c = 10^2$

a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

b) Determine the form of compensator required to achieve the desired crossover frequency and phase margin. Lead Compensator

c) Determine the desired phase lead/lag that the compensator must provide at the crossover frequency ω_c. to achieve 45° phase margin we need to increase phase about 57.97°



Example

Hints:

The maximum phase shift of a lead compensator occurs at $\omega = \sqrt{(ab)}$

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

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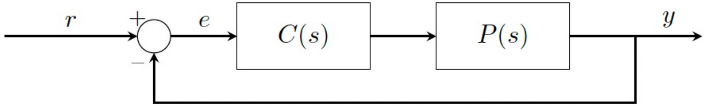

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a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

b) Determine the form of compensator required to achieve the desired crossover frequency and phase margin. Lead Compensator

c) Determine the desired phase lead/lag that the compensator must provide at the crossover frequency ω_c. 57.97°

d) At what point do we want the maximum phase shift of our compensator to occur? Use this to formulate a requirement on the parameters a and b. Combine it with your result from (c) to determine a and b.

$57.97° = 2\arctan\left(\sqrt{b/a}\right) - 90°$

① $147.97° = 2\arctan\sqrt{b/a}$

② $\sqrt{ab} = 100$

solve

$\rightarrow a \approx 29{,}67 \quad 348$

Example

Design a lead-lag compensator $C(s)$ for a system modeled by Figure 3 with

$$P(s) = 5\frac{(1-\frac{s}{200})}{(1+\frac{s}{20})(1+\frac{s}{4})}$$

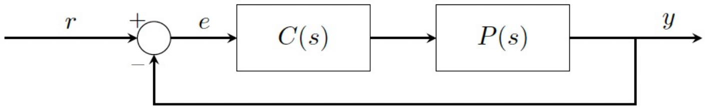

Figure 3: Standard feedback architecture.

In order to design the compensator, the problem has been split into the following subtasks:

3.1 First, we look to ensure that the rise time is as short as possible while maintaining a phase margin in the range from 40° to 45°.

a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

$\omega_c = 10^2$

e) Finally, by relying on the definition of the crossover frequency, determine a suitable value for the gain K of the compensator.

$$K|P(j\omega_c)| \overset{!}{=} 1$$

$$P(100j) = 5\frac{(1-\frac{100j}{200})}{(1+\frac{100j}{20})(1+\frac{100j}{4})} = 5\frac{(1-0.5j)}{(1+5j)(1+25j)} = 5\frac{\sqrt{1^2+0.5^2}}{\sqrt{1+5^2}\sqrt{1+25^2}}$$

$$K = \frac{1}{|P(100j)|} \approx 6.56$$

Example

$$C_{lead} = \frac{s/a + 1}{s/b + 1}, 0 < a < b$$

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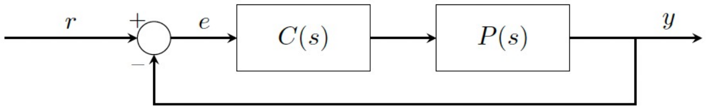

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a) Using Hints 1 and 2 above, determine a suitable crossover frequency ω_c for the system.

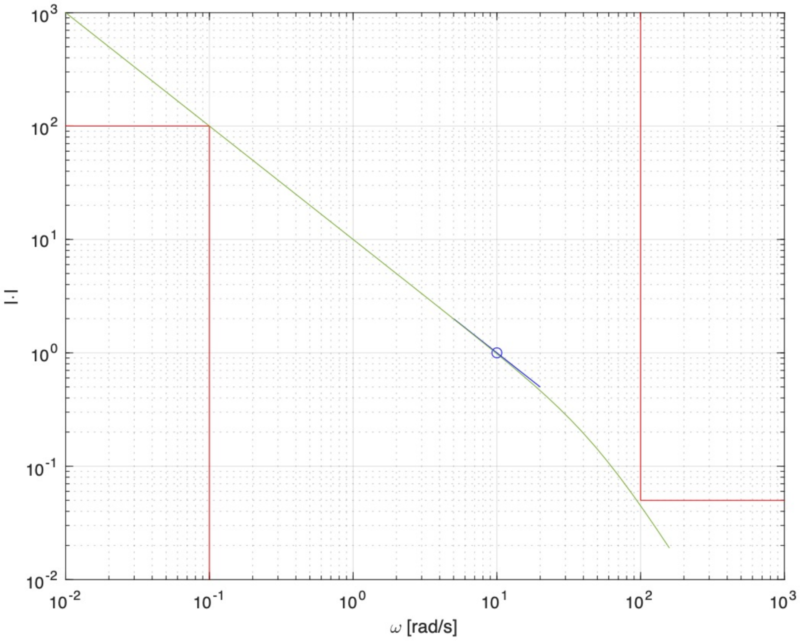
e) Finally, by relying on the definition of the crossover frequency, determine a suitable value for the gain K of the compensator. $K = 6.56$ $a \approx 29 \; \& \; b = 348$

putting everything together we get:

$$C_{\text{lead}}(s) = 6.56\frac{1 + s/29}{1 + s/348}$$

Performance Limitations

- The requirement to keep a shallow slope over a "large" interval near crossover may make it impossible to clear the obstacles
- nmn-phase zeros and unstable OL poles may make the situation even harder
- Many performance goals conflict, e.g., you cannot achieve strong disturbance rejection, noise attenuation, and good tracking at the same time



Tips for Problem Set 10

Easy	Medium	Hard
	$1, 2, 3$	4

1: a) look up the graphical definitions in the slides. c) think about negative & positive margins. d) recall the Nyquist stability criterion

2: a) $|kL(j\omega_{gc})| = 1$
b) look at the mathematical definitions for phase & gain margin and plug in the open-loop TF

3: a) we did it graphically together for intuition, try to do the calculations if you wanna get the exact results b) follow the tips and remember $e_{ss} = \lim_{s\to 0} \frac{1}{1+L(s)}$
for small ω $L(s)$ large, so $\lim_{s\to 0} \frac{1}{1+L(s)} = \frac{1}{L(s)}$
which compensator do we choose to increase magnitude?

4.1: we need atleast as many poles @ origin, as order of ramp b) remember the table summarizing e_{ss} or use FVT
4.2: use the definition of sensitivity function to calculate the error
4.3: remember how we changed ω_{gc} in our example by multiplying the TF with a gain K

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header